

LAERDAL IV, Norway

WMO: 013550

Lat: 61.103N

Lon: 7.503E

Elev: 5

StdP: 101.26

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -12.5	(c) -10.4	(d) -17.6	(e) 0.8	(f) -7.0	(g) -15.9	(h) 0.9	(i) -8.3	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.8	(c) 25.4	(d) 16.1	(e) 23.3	(f) 15.1	(g) 21.6	(h) 14.4	(i) 17.5	(j) 22.8	(k) 16.6	(l) 21.0	(m) 15.8	(n) 19.9	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	11.1	18.5	14.8	10.5	17.8	14.0	10.0	16.8	49.2	22.3	46.5	21.1	44.1	20.1	20.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) -12.7	(f) 28.7	(g) 2.5	(h) 3.5	(i) -14.5	(j) 31.2	(k) -16.0	(l) 33.2	(m) -17.3	(n) 35.1	(o) -19.1	(p) 37.7		
(4) N/A	(5) N/A	(6) N/A	DB	WB	-13.1	18.9	2.2	0.9	-14.7	19.5	-16.0	20.0	-17.2	20.6	-18.8	21.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-0.9	-0.7	2.8	6.7	10.7	13.9	17.1	15.1	11.5	6.0	2.4	0.0
	DBStd	7.12	4.35	4.68	3.45	2.87	2.98	2.25	2.92	2.23	2.78	3.83	4.11	4.76
	HDD10.0	1681	338	299	223	106	27	2	0	0	16	133	228	309
	HDD18.3	4127	596	532	482	350	238	132	60	104	205	382	477	568
	CDD10.0	621	0	0	0	6	48	120	219	157	61	9	1	0
	CDD18.3	24	0	0	0	0	0	1	21	3	0	0	0	0
	CDH23.3	184	0	0	0	0	3	11	154	16	0	0	0	0
	CDH26.7	32	0	0	0	0	0	0	32	0	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	688	69	58	58	31	33	48	57	62	60	64	64	72
	PrecMax	1008	201	155	262	69	60	133	107	107	101	169	170	179
	PrecMin	464	0	0	4	9	8	13	19	14	20	19	1	7
	PrecStd	125	55	44	52	17	14	30	21	25	24	35	45	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	10.7	12.4	18.2	23.4	24.7	29.8	25.2	20.4	18.5	14.1	11.0
		MCWB	4.4	5.2	7.1	9.8	13.7	14.1	17.7	14.6	12.8	11.6	8.3	5.5
	2%	DB	7.9	8.8	10.2	16.0	21.1	22.5	27.0	23.0	18.6	15.7	11.8	8.8
		MCWB	3.6	3.8	5.4	9.0	12.7	13.0	17.3	14.9	12.5	10.2	6.7	4.3
	5%	DB	5.4	7.2	8.9	13.9	18.8	20.8	24.8	21.3	17.2	12.8	9.7	6.9
		MCWB	2.3	2.7	4.4	7.6	11.7	12.6	16.4	14.6	12.3	8.9	5.8	3.0
	10%	DB	4.1	5.7	7.6	11.8	16.5	19.2	22.8	19.6	16.0	11.1	7.6	5.8
		MCWB	1.3	2.0	3.9	6.5	10.5	12.1	15.7	14.2	11.6	7.7	5.3	2.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	6.1	7.8	10.9	15.1	15.9	19.2	17.5	15.3	12.5	9.2	6.5
		MCDB	7.4	8.8	11.0	16.8	21.7	21.5	26.4	21.4	17.2	15.5	12.4	9.1
	2%	WB	4.2	4.5	6.5	9.5	13.5	14.6	18.1	16.5	14.1	11.2	8.0	5.2
		MCDB	6.5	7.2	9.0	15.0	19.2	19.3	24.5	19.9	16.3	15.1	11.1	7.4
	5%	WB	3.3	3.4	5.4	8.3	12.3	13.8	17.3	15.8	13.3	9.8	6.7	4.3
		MCDB	5.5	6.4	7.8	12.9	17.4	17.8	22.5	19.1	15.6	12.3	8.7	6.6
	10%	WB	2.0	2.5	4.4	7.2	11.3	13.1	16.6	15.0	12.6	8.4	5.5	3.3
		MCDB	3.5	5.2	6.9	10.9	15.8	17.0	21.3	18.2	15.0	10.5	7.6	5.3
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.9	6.3	8.2	9.2	9.2	8.8	7.7	6.7	5.5	4.8	4.5
		MCDBR	6.0	6.2	7.2	11.4	13.4	12.6	12.9	10.4	8.0	7.1	7.3	6.6
		MCWBR	3.5	3.4	4.1	5.8	6.1	5.4	4.7	4.5	3.8	3.7	4.0	3.9
	5% WB	MCDBR	5.3	5.8	6.8	10.2	12.2	9.7	11.1	8.7	7.2	6.2	7.0	6.6
		MCWBR	3.4	3.6	4.3	5.4	5.8	4.7	4.5	4.2	3.9	3.7	4.1	4.4
Clear-Sky Solar Irradiance	taub	0.234	0.280	0.318	0.354	0.360	0.357	0.373	0.364	0.330	0.305	0.244	0.203	
	taud	2.083	2.189	2.137	2.105	2.200	2.354	2.379	2.398	2.472	2.367	2.119	1.976	
	Ebn at Noon	498	698	788	828	858	869	844	821	791	671	471	351	
	Edh at Noon	39	68	101	127	126	110	105	95	74	58	37	25	
All-Sky Solar Radiation	RadAvg	0.22	0.73	1.96	3.72	4.79	4.84	4.61	3.54	2.21	1.01	0.32	0.14	
	RadStd	0.05	0.21	0.40	0.63	0.74	0.54	0.53	0.35	0.28	0.15	0.05	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page